

Finlay Cooper

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Education

Western University, London ON

June 2026

Bachelor of Engineering, Software Engineering | **Dean's Honour List**

- **Core Coursework:** Algorithms & Data Structures; Operating Systems; DBMS; Networking; Web Technologies; Information Security; Machine Learning; Cloud Computing; Software Testing & Maintenance.

Experience

Software Developer Intern, BroadGrain Commodities — Toronto, ON

Summer 2025

- **Owned full product lifecycle** as the sole developer—defined requirements with plant managers, managed backlog, and shipped iterative releases under real operational deadlines.
- Engineered a **schema-driven backend factory** to auto-generate CRUD controllers, routes, and validation from a single schema—reducing 100+ files to 3 and enabling zero-code endpoint creation.
- Integrated **Microsoft Graph API + Azure AD** to automate SharePoint document delivery with token caching, retries, and exponential backoff—eliminating manual report distribution.
- Built **10 PDF reporting services** (JSReport, Handlebars, digital signatures, multi-photo support), replacing paper-based compliance workflows across 8 categories.
- Deployed to on-prem infrastructure (**pm2/IIS**); authored runbooks and onboarded successor for system handoff.
- **Reduced processing time by 15% (refactoring) and 75% (digitization)** across reporting workflows.

Projects

GTA Management System — Capstone | control-alt-elites.netlify.app

2025–2026

- Built a **full-stack platform** replacing spreadsheet-based GTA hiring at Western Engineering—**deployed for department-wide use** across 100+ TAs, professors, and committee members.
- Designed a **configurable matching algorithm** combining skill match, preferences, course history, and review data with adjustable weights—executing assignments across 4 priority waves.
- Developed **78+ REST API endpoints** with role-based access control (student/professor/committee), Zod validation, and ownership enforcement—fully tested via automated bash suite.
- Implemented a **dry-run assignment system** enabling preview, weight tuning, and manual overrides before commit—improving transparency and decision control.

Soi Ram — Freelance Restaurant Website | soiram.netlify.app

2025–2026

- Delivered a **production-grade Next.js 16 / TypeScript / Tailwind site** with dynamic theming, fluid `clamp()` typography, and data-driven menu rendering—managed full client lifecycle and deployment.
- Integrated **Toast POS ordering** alongside UberEats and DoorDash—enabling commission-free direct orders with a unified receipt-style UI.
- Designed a **retro 80s brand identity** with custom typography, mobile-first layout, and SEO optimization.

CloudCopier — Cross-Device Clipboard Sync | github.com/FinlayCooper/CloudCopier

2024–2025

- Implemented **native clipboard monitoring** on Windows (P/Invoke, `SetClipboardViewer`) and iOS (`UIPasteboard`) with timestamp-based conflict resolution to prevent sync loops.
- Built a **cross-platform sync system** (C#, Swift, Node.js) supporting text, images, and 40+ file types (50 MB)—containerized with Docker and deployed with auto-restart health checks.
- Engineered **format normalization, deduplication, and retry logic** for reliable synchronization; set up CI/CD for automated builds and deployments.

Skills

Languages & Frameworks: TypeScript/JavaScript, Java, C#, Python, C++, SQL; React/React Native, Next.js, Node.js, Express.js, Swift/SwiftUI, Unity

Tools & Platforms: MySQL, Firestore, PostgreSQL, MongoDB; Prisma ORM; Azure/SharePoint, IIS, Nginx, Docker, Git, CI/CD; Networking (IP, Routing & Switching)

Spoken Languages: English, French